

PERFORMANCE SPECIFICATION
FOR
EXTENDABLE BOOM FORKLIFT

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1.0 SCOPE

1.1. Scope. This specification prescribes the performance requirements and identifies verification procedures for the Extendable Boom Forklift (EBFL). The United States Marine Corps (USMC) has a requirement to replace the EBFL

1.2. Requirement Levels. This specification lists two values for certain performance parameters. The threshold (T) is the minimum acceptable level. The objective (O) is the desired level at which performance of the EBFL results in an operationally significant increase in capabilities. When only one requirement is stated, it is the threshold requirement.

2.0 APPLICABLE DOCUMENTS

2.1. General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether they are listed.

2.2. Government documents

2.2.1. Government specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130N(1)	Identification Marking of US Military Property
MIL-STD-1366E	Transportability Criteria
MIL-STD-209K	Interface Standard for Lifting and Tiedown Provisions
MIL-STD-461G	Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment
MIL-STD-1179F	Lamps, Reflectors and Associated Signaling Equipment for Military Vehicles
MIL-PRF-10924H(2)	Grease, Automotive and Artillery
MIL-PRF-11021J	Switch, Vehicular Lights: 28 Volt DC
MIL-PRF-2104N	Lubricating Oil, Internal Combustion Engine, Combat/Tactical Service
MIL-PRF-23827C(2)	Performance Specification, Grease, Aircraft and Instrument, Gear and Actuator Screw
MIL-PRF-46170E	Performance Specification Hydraulic Fluid, Rust Inhibited, Fire Resistant, Synthetic Hydrocarbon Base, NATO Code No. H-544

Performance Specification for Extendable Boom Forklift

MIL-PRF-62122E Performance Specification Cable Assembly, Inter-Vehicle Power: Plug, Receptacle, and Adapter

SOCIETY OF AUTOMOTIVE ENGINEERS STANDARDS

SAE-AMS-M-7866 Molybdenum Disulfide, Technical, Lubrication Grade

SAE-AMS-STD-595 Colors Used in Government Procurement

SAE J88 Machines, Off-Road Work, – Exterior, Sound Measurement

SAE J551 Performance Levels and Methods of Measure of Electromagnetic Compatibility of Vehicles, Boats, and Machines

SAE J595 Directional Flashing Optical Warning Devices for Authorized Emergency, Maintenance, and Service Vehicles

SAE J845 Optical Warning Devices for Authorized Emergency, Maintenance, and Service Vehicles

SAE J994 Alarm, Backup, Electric Laboratory Performance Testing

SAE J919 Sound Measurement - Off-Road Work Machines - Operator - Singular Type

SAE J1127 Low Voltage Battery Cable

SAE J1166 Sound Measurement—Off-Road Self-Propelled Work Machines Operator-Work Cycle

SAE J1503 Performance Test for Air-Conditioned, Heated, and Ventilated Off-Road, Self-Propelled Work Machines

SAE J1939 Serial Control and Communications Heavy-Duty Vehicle Network

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION

ISO 3471 International Organization for Standardization Earth moving Machinery Falling Object Protective Structures.

NATO-AEP-3.1.12.1.5 Military Load Classification of Bridges, Ferries, Rafts, and Vehicles

NATO STANAG 2021 Military Load Classification of Bridges, Ferries, Rafts, and Vehicles

NATO STANAG 4074 Auxiliary Power Unit Connections for Starting Tactical Land Vehicles

HANDBOOKS FOR MILITARY

Army FM 3-11.5 CBRN Decontamination Multiservice Tactics, Techniques, and Procedures for Chemical, Biological, Radiological, and Nuclear Decontamination

Marine Corps Order (MCO) 4790.25 Ground Equipment Maintenance Program (GEMP)

Natick TR-13/018 2010 Anthropometric Survey of US Marine Corps Personnel: Methods and Summary Statistics

OTHER STANDARDS

Federal Motor Vehicle
Safety Standard (MVSS)
108

Lamps, Reflective Devices, and Associated Equipment

OSHA 1910.1200

Occupational Safety and Health Standards for Hazard Communication

2.3. Order of Precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption is obtained.

3.0 REQUIREMENTS

3.1. Fork Attachments. The EBFL shall be equipped with one or multiple fork attachments that can meet the requirements below.

3.1.1. Container Operations. The EBFL shall be capable of loading and unloading standard military pallets (40 inches by 48 inches) from 20' International Organization for Standardization (ISO 8 feet x 8 feet x 20 feet) container on the ground and raised 55 inches. The pallets will be loaded in two rows longitudinally. This may be achieved by a fork attachment specific to this application.

3.1.2. CH-53K Cargo Operations. The EBFL shall be capable of loading and unloading a longitudinally orientated 463L pallet (88 inches x 108 inches) weight up to 10,000 pounds with a 54-inch load center into the rear of the CH-53K. This may be achieved by a fork attachment specific to this application.

3.1.3. Retracted Lift Capacity. The EBFL shall be capable of lifting and traveling with a 10,000-pound load with a 54-inch load center.

3.1.4. Sling Load. The EBFL boom shall be equipped with a provision (i.e. hook, eyelet, etc.) for sling lifting material.

3.1.5. Horizontal Extended Lift Capacity

3.1.5.1. 20 feet Extension. The EBFL shall be capable of lifting a minimum load of at least 4,000 pounds loaded on a standard military pallet (40 inches by 48 inches) at a 24-inch load center at a minimum horizontal boom extension of 20 ft.

3.1.5.2. 15 feet Extension. The EBFL shall be capable of lifting a minimum load of at least 6,000 pounds at a horizontal boom extension of 15 ft.

3.1.6. Safe Working Load. The EBFL shall be certified to lift at least 1.5 times the rated load at the applicable load center without the rear wheels leaving the ground.

3.1.7. Fork Carriage. If two carriages are required, each fork carriage shall meet the requirements for size, tilt, oscillation, offset, shift, and spacing.

3.1.7.1. Fork Tines (Length). The EBFL fork tines shall be long enough to engage a 108 inch long 463L pallet.

3.1.7.2. Fork Tines (USAF Pallet Forks). The EBFL fork tines shall be compatible with the USAF Conveyorized Pallet forks.

3.1.7.3. Tilt. The Fork Carriage shall tilt at least 10 degrees forward and 20 degrees rearward.

3.1.7.4. Oscillating. The Fork Attachment shall oscillate at least 6 degrees to each side of center.

3.1.7.5. Offset/Shift. The Fork Attachment shall offset or shift the forks laterally to either side of the carriage at least 6 inches.

3.1.7.6. Fork Tine Spacing. The Fork Attachment shall be capable of varying the distance between fork tines from 6 to 72 inches.

3.1.8. Load Chart. The EBFL shall provide a load chart identifying the system lifting capabilities and restrictions in accordance with the markings requirements in ANSI B56.6. The chart shall be visible to the operator in the operating position. The chart shall correspond with boom elevation and extension indicators.

3.2. Maneuverability

3.2.1. Steering Modes. The EBFL shall be capable of two-wheel, four-wheel, and crab steering. An in-the cab indicator shall show, at a glance, the position of the rear wheels.

3.2.2. Terrain. The EBFL shall be capable of traveling with and without rated load and conducting load handling operations on both improved and unimproved surfaces.

3.2.3. Service Brake. The service brakes shall be capable of stopping and holding the EBFL, with rated load, while backing down a 45 percent (24.23 degree) slope.

3.2.4. Parking Brake. The parking brake shall restrain the EBFL, with rated load, from movement on a 20 percent (11.31 degree) slope.

3.2.5. Longitudinal Slope. The EBFL, with rated load, shall be capable of negotiating an approach and departure angle up to 20 degrees.

3.2.6. Side Slope. The EBFL, with rated load, shall be capable of negotiating up to a 30 percent cross slope in full circle operation in both directions at maximum steer angle without any tire leaving the ground.

3.2.7. Fording. The EBFL shall be capable of fording at least 30 inches (Threshold) or 60 inches (Objective) of salt water with a fording kit. The fording requirement includes wave action.

3.2.8. Speed. The EBFL shall be capable of traveling at 15 miles per hour on flat firm ground with rated load.

3.2.9. Turning Radius. The EBFL shall be capable of executing a 360 degree turn within a curb-clearance circle of 35 feet, maximum.

3.3. Physical Characteristics

3.3.1. Height (Cab). The EBFL shall be capable of lowering and operating at a cab height of 84 inches. The EBFL is desired to maintain a cab height of 84 inches.

3.3.2. Maximum Height (CH-53K Loading). The EBFL, with rated load while loading a CH-53K, shall be capable of loading a deck height of 31 inches and no part of the vehicle shall exceed a maximum height of 84 inches.

3.3.3. Width. The EBFL shall have an overall width not to exceed 102 inches, including tire bulge.

3.3.4. Length. The overall EBFL length shall not exceed 298 inches.

3.4. Transportability

3.4.1. Internal Air Transportability. The EBFL shall be air transportable inside a C-130, C-17, and C-5 aircraft with forks and counterweight in accordance with MIL-STD-1366E. If any assembly/disassembly to or from the internal transport configuration is required, procedures shall not exceed two man-hours, use the integral lifting devices, be completed with common tools, and removed components shall be transportable in the same aircraft.

3.4.2. External Air Transportability. The EBFL shall be externally lifted by the Ch-53K in accordance with MIL-STD-913A.

3.4.3. Highway Transport. The EBFL shall be capable of unrestricted transportation on highways both as a standalone vehicle as Gross Vehicle Weight and as cargo on a trailer at Vehicle Cargo Weight in accordance with MIL-STD-1366E.

3.4.4. Rail Transport. The EBFL shall be capable of unrestricted transportation by railcar in accordance with MIL-STD-1366E.

3.4.5. Marine Transport. The EBFL shall be capable of unrestricted marine transportation without any disassembly in accordance with MIL-STD-1366E.

3.4.6. Roll-On/Roll-Off. The EBFL, without rated load, shall be capable of negotiating the ramp of a landing craft at a longitudinal slope up to 45 percent (24.23 degrees) slope in forward and reverse gears.

3.4.7. Lifting and Tiedown Provisions. The EBFL lifting and tiedown provisions shall be certified to the MIL-STD_209K, Type 1 and MIL-STD-913A. The provisions shall be labeled "LIFT", "TIEDOWN", or "LIFT TIEDOWN", as appropriate, in one-inch high letters.

3.4.8. Towing Pintle. The EBFL shall be equipped with a towing pintle located on the rear of the machine, capable of moving equivalent to the EBFL weight and a tongue weight approximately 10 percent of the EBFL weight.

3.5. Operator's Cab

3.5.1. Visibility. The visibility from the EBFL operator's cab shall enable the operator to observe the fork tines when engaging the tineways of a container or pallet. The cab shall be equipped with commercial features to include at a minimum: defroster, mirrors, and electronically operated windshield wipers.

3.5.2. Human Factors Engineering. The EBFL shall be maintainable and operable by 5th percentile female through 95th percentile male Marine per Natick TR-13/018, dressed in environmental protective clothing.

3.5.3. Personal Protection Equipment. The EBFL shall be operable by the 5th percentile female to the 95th percentile male Marine per Natick TR-13/018 in combat gear.

3.5.4. Cold Weather Gear. The EBFL shall be operable by the 5th percentile female to the 95th percentile male Marine per Natick TR-13/018 while wearing Cold Weather Gear.

3.5.5. Chemical, Biological, Radiological, Nuclear. The EBFL shall be operable by the 5th percentile female to the 95th percentile male Marine per Natick TR-13/018 while wearing full CBRN Protective clothing Mission Oriented Protective Posture (MOPP) four.

3.5.6. Air Conditioner. The air conditioning system shall be capable of operating in cooling mode at ambient temperature between 65 degrees Fahrenheit and 125 degrees Fahrenheit and shall meet the cooling requirements of SAE J1503. The system shall use an Environmental Protection Agency (EPA) compliant refrigerant.

3.5.7. Heater. The heater shall be capable of warming the cab from -25 degrees Fahrenheit to +41 degrees Fahrenheit within an hour after starting the EBFL when the exterior ambient temperature is -25 degrees Fahrenheit. The heater shall meet the heating requirements of SAE J1503.

3.5.8. Instrumentation. The instrument panel shall be protected from weather elements.

3.5.9. Horn. The EBFL shall be equipped with an electric or air horn.

3.5.10. Backup Alarm. The EBFL shall be equipped with an audible back up alarm conforming to SAE J994 with a minimum of 10 dBA greater than the surrounding background noise of the operating loader.

3.5.11. Ignition. A single keyless engine switch shall be provided in the cab. The switch shall control all electrical monitors and electrical equipment, except the horn and lights.

3.5.12. Safety. The EBFL shall comply with current safety, health and ergonomic standards in accordance with American Society of Mechanical Engineers (ASME), Society of Automotive Engineers (SAE), American National Standards Institute (ANSI), Environmental Protection Agency (EPA), and Occupational Safety Health Administration (OSHA) as standards guidelines.

3.5.13. Rifle Mount. The EBFL Operator's cab shall be equipped with a universal Rifle Mount capable of holding the M4 and M16A4. The rifle will have a magazine inserted and accessories such as the Rifle Combat Optic (RCO), AN/PEQ-15, and suppressor. The rifle, while mounted in the retention system, shall be located away from the operator's controls, not obstruct the vision of the operator, and be within arm's reach of the operator. The system shall securely hold the rifle and prevent damage or accidental discharge during forklift operations.

3.5.14. Noise Limits. The EBFL interior cab noise shall be below 85 Decibel A when measured in accordance with SAE J1166 and J919.

3.5.15. Hazardous Materials. The EBFL shall not contain Class I or Class II ozone-depleting substances, asbestos, cadmium, hexavalent chromium, or highly toxic or carcinogenic materials as defined in OSHA 1910.1200 without government approval.

3.5.16. Fire Extinguisher. The EBFL operator's cab shall be equipped with a fire extinguisher with a minimum rating of 10B: C (i.e. NSN 4210-00-775-0127). The extinguisher shall be mounted securely in the operator compartment.

3.5.17. Roll-Over Protection Structures (ROPS). The EBFL shall be equipped with ROPS conforming to ISO 3471.

3.5.18. Falling-Object Protective Structures (FOPS). The EBFL shall be equipped with FOPS conforming to ISO 3471.

3.6. Tires

3.6.1. Tires. The tires shall be sufficient to meet the lifting and weight requirements of the EBFL.

3.6.2. Onboard Air Compressor. The EBFL shall be equipped with an engine driven air compressor for inflating the tires. The system shall include all tools and adequate length hose necessary to fill all tires.

3.6.2.1. Pneumatic Tool Storage. The EBFL pneumatic tools shall be stored on board in a storage container.

3.6.3. Offroad and Beach Operations. The tires shall enable the EBFL, with rated load, to operate in various offroad and beach conditions.

3.7. Paint

3.7.1. Corrosion Protection. The EBFL shall be resistant to corrosion during unprotected storage and from the effects of a saltwater environment.

3.7.2. Paint and Treatment. The EBFL must be painted green 383 in accordance with SAE-AMS-STD-595.

3.7.3. Operational Decontamination. The EBFL shall withstand deterioration from operational decontamination as specified in Army Field Manual (FM) 3-11.5 Chapter IV.

3.8. Environmental

3.8.1. Low Ambient Temperature (Start-up). The EBFL shall be capable of starting unassisted within 5 minutes of initial attempt in ambient temperatures as low as -25 degrees Fahrenheit.

3.8.2. Low Ambient Temperature (Operation). The EBFL shall be capable of operating with a smooth-running engine within 15 minutes of start up in ambient temperatures as low as -25 degrees Fahrenheit.

3.8.3. High Ambient Temperature. The EBFL shall be capable of starting and operating in ambient temperatures as high as 125 degrees Fahrenheit.

3.9. Lighting

3.9.1. Work Lights. The EBFL shall be equipped with commercial work lights, that are brush-guard protected, adjustable, and located to illuminate the work area for night operations.

3.9.2. Road Lights. The EBFL shall be equipped with lighting for road travel in accordance with FMVSS 108 and SAE J595 and SAE J845. The EBFL shall be equipped Headlights, taillights, Brake lights, Reverse lights, Indicator Lights.

3.9.3. Blackout Lighting. The EBFL shall be equipped with one head and two tail/stop blackout lighting system conforming to MIL-STD-1179F. An interior switch conforming to MIL-PRF-11021J, which is readily accessible to the operator in the seated position, shall control the system. All other exterior lights, backup alarm and the horn, if electric, shall be made inoperable automatically when the blackout lights are switched on. The blackout mode switch shall activate all blackout lights and markers.

3.10. Electrical System

3.10.1. 24V Electrical System. The EBFL shall be equipped with 12/24 dual system with a 24-Volt alternator, 24-Volt Starter, and battery equalizer (Threshold), or a 24-Volt negative-ground electrical system (Objective).

3.10.2. Batteries. The EBFL shall be equipped with the proper quantity of lead acid absorbed glass-mat (AGM) batteries to support the system. The EBFL shall be compatible with Hawker ARMSAFE batteries NSN 6140-01-485-1472.

3.10.2.1. Battery Accessibility. The EBFL batteries shall be accessible to personnel standing on the ground for removal without restriction and labeled with one-inch black lettering.

3.10.2.2. Battery Disconnect Switch. A master switch shall be provided that will disconnect the batteries from the electrical system thus preventing accidental draining of the batteries when the EBFL is not in use. The switch shall be accessible and conveniently located for operation.

3.10.2.3. Battery Cables. The EBFL shall be equipped with battery cables with insulated covers and shall conform to SAE J1127. Corrosion resistant bolts and nuts shall be provided.

3.10.3. NATO Receptacle. The EBFL shall be equipped with a NATO Receptacle capable of input and output electrical current from another NATO Receptacle conforming to NATO STANAG-4074, Type 1 and in accordance with MIL-PRF-62122E. The receptacle shall be accessible to the personnel standing on the ground and labeled with one-inch black lettering.

3.10.4. Electromagnetic Interface

3.10.4.1. Electromagnetic Interface. The EBFL shall meet the requirement of MIL-STD-461G for radiated emissions and susceptibility.

3.10.4.2. Electromagnetic Interface. The EBFL shall meet the requirements of RE 102 for Navy mobile and Army ground-based systems.

3.10.4.3. Electromagnetic Interface. The EBFL shall meet the requirements of RS 103 for ground-based systems of 50 V/m over the frequency range of 2 MHz to 18 GHz.

3.11. Drivetrain

3.11.1. Transmission. The EBFL transmission shall be a automatic transmission that is commercially available to construction industry and not unique to the USMC application.

3.11.1.1. Transmission Oil Sample Valve. The EBFL shall have an oil drain petcock sample valve located at the lowest point of the tank or hydraulic system.

3.11.1.2. Four Wheel Drive. The EBFL shall be equipped with four-wheel drive capability.

3.11.2. Engine. The EBFL shall be equipped with an engine capable of operating on diesel, JP-5, JP-8, and F-24 fuel types.

3.11.2.1. Fuel Capacity. The EBFL shall be capable of completing two missions without refueling. One mission is defined as 4 hours (1.8 hours traveling, 2.2 hours lifting)

3.11.2.2. Engine Oil Sampling Valve. The EBFL engine shall have an oil drain petcock sample valve.

3.11.2.3. Engine Oil Drain. The EBFL shall have an oil change system for quick service of the engine.

3.11.3. Exhaust. The EBFL exhaust system shall be capable of mounting a spark arrestor (unless turbo-charged) and be equipped with a cap that restricts rain from entering the exhaust.

3.11.4. Hydraulic System

3.11.4.1. Hydraulic Fluid Sample Valve. The EBFL shall have a drain petcock valve located at the lowest point of the tank or hydraulic system.

3.11.5. Military Petroleum, Oil, and Lubricants (POL). The EBFL shall be capable of accepting military standard petroleum, oil and lubricants (POL).

a. Engine: MIL-PRF-2104N

b. Hydraulic systems: MIL-PRF-46170

c. General Lubrication: MIL-PRF-10924, SAE-AMS-M-7866, and MIL-PRF-23827C (when cold starting kit is required)

3.12. Maintainability.

3.12.1. Maintenance Infrastructure. The corrective and preventive maintenance must be supportable within the current two levels of maintenance infrastructure in accordance with Marine Corps Order (MCO) 4790.25.

3.12.2. Preventive Maintenance. Scheduled preventive maintenance, except operational checks, shall be required not more often than every 50 hours. Before/during/after operational checks will be performed in accordance with the manufacturer's manual.

3.12.3. Onboard Diagnostics

3.12.3.1. Commercial Diagnostic Systems. The EBFL onboard diagnostic systems shall connect with standard commercial diagnostic tools and equipment with a CAN Bus 9-pin connector in accordance with SAE J1939. (T)

3.12.3.2. USMC Diagnostic Systems. The EBFL onboard diagnostic system is desired to be compatible with Wireless At Platform Test System (WATS), and Electronic Maintenance Support System (EMSS) (O).

3.12.3.3. Diagnostic Capabilities (Read, Write, Commands). The EBFL diagnostic system shall allow maintainers to perform read, write, and bi-directional commands, including fault code retrieval, system status monitoring, calibration, resets, and functional tests.

3.12.3.4. Diagnostic Capabilities (License). The EBFL diagnostic system shall be unrestricted, without reliance on OEM-only tools/licensing.

3.12.3.5. Diagnostic Capabilities (Documentation). The EBFL diagnostic system shall allow Government access to all documentation, protocols, and software for all echelons of maintenance.

3.12.4. RAM (MTBF). The minimum acceptable value (MAV) over the life of the EBFL shall be at least 250 hours mean time between failure (MTBF) with a confidence level of 85 percent and with no more than a 0.25 maintenance ration (MR). The expected length of a mission is 4 hours (1.8 hours for travel and 2.2 hours for lifting). A failure is defined as any problem that prevents the machine from traveling and lifting the rated load.

3.12.5. RAM (MTTR). The mean time to repair (MTTR) shall be no more than 2.5 hours at the organizational maintenance level and 3.5 hours at the intermediate maintenance level. The maximum time to repair at each of the above echelons of maintenance shall be no more than 4.5 and 8.0 hours, respectively.

3.12.6. Rebuild/Replace. The EBFL shall be capable of completing at least 2,000 hours of operation before replacement or rebuild of major components, such as the engine, transmission, power train, or hydraulic system.

3.13. Identification

3.13.1. Identification and Markings. Each EBFL shall be furnished with an identification plate showing the specific UID markings as defined in MIL-STD-130N(1). The identification plate shall be permanently affixed to the EBFL in a readily accessible and conspicuous location.

3.13.2. Instruction Plates. The EBFL shall be equipped with instruction plates, or diagram plates, describing procedures to be followed for operating, servicing, and lubrication. The plates shall be permanently affixed to the EBFL in a location that's protected and conspicuous.

3.13.3. Military Load Classification Identification. A Military Load Classification number for the loader shall be displayed on the EBFL in accordance with NATO-STANAG 2021 and NATO-AEP-3.12.1.5. The number shall be 3.5-inch gothic black numerals centered within a 9-inch diameter, 1/4-inch-thick black border displayed on the front (curbside) of the vehicle and below the driver's line of vision. The number may be painted on the vehicle or placed on a plate permanently mounted to the vehicle. The number shall be visible at ground level at a distance of 20 feet from the left front of the vehicle. The government will assign the Military Load Classification number.

4.0 VERIFICATION

4.1. General. Compliance with the requirements listed in Section 3 shall be verified using the methods established in Section 4. Refer to Table 4 for a summary of the verification method.

4.2. Methods of Verification

4.2.1. Certification (C). The Contractor shall provide independent certification that the EBFL complies with all specific operating, operating environment, interface, ownership, and support requirements stated within this performance specification.

4.2.2. Analysis (A). Analyses shall be conducted by the Contractor as specified herein or the Government utilizing established technical or mathematical models or simulations, algorithms, charts, graphs, circuit diagrams, or other scientific principles and procedures.

4.2.3. Inspection (I). Inspections shall be conducted by the Contractor as specified herein; or by the Government at a Government-designated test agency or by a Quality Assurance Representative (QAR) at the Contractor's facility utilizing sight, hearing, smell, or touch; simple physical manipulation; and mechanical and electrical gauging and measurement.

4.2.4. Demonstration (D). Demonstrations shall be conducted by the Contractor as specified herein; or the Government at a Government-designated test agency involving the actual operation of the system to provide evidence that the required functions were accomplished under specific scenarios. The system may be instrumented, and performance monitored.

4.2.5. Test (T). Tests shall be conducted by the Contractor as specified herein; or by the Government at a Government-designated test agency applying scientific principles and procedures to determine the properties or functional capabilities of the system.

4.3. Responsibility for compliance. The absence of any verification method within this performance specification shall not relieve the Contractor of the responsibility for acceptance and compliance. Unless otherwise specified within the contract or performance specification, the Contractor is responsible for the execution of all inspections and certifications as stated in this performance specification. The Contractor is responsible for ensuring that components and materials are manufactured, inspected, and tested in accordance with referenced specifications and standards, as applicable. Failure to comply with any of the requirements shall be cause for rejection.

4.4. Conformance inspections/tests. EBFLs that have met all inspections, test, and certification requirements at the factory, shall be shipped to a Government site. The Government reserves the right to validate requirement conformance through independent analyses, inspections and/or testing. Deficiencies found because of testing/inspection shall be a cause for rejection until the Contractor has provided evidence that corrective action has been taken to eliminate the deficiency. The Contractor, at no additional cost to the Government, shall correct such deficiencies.

Table 4. Requirement/Verification Cross Reference Matrix

Paragraph	Title	Method				
		C	A	I	D	T
3.1	Fork Attachments	X				
3.1.1	Container Operations	X				
3.1.2	CH-53K Cargo Operations					X
3.1.3	Retracted Lift Capability					X
3.1.4	Sling Load					X
3.1.5.1	Horizontal Extended Lift Capacity (20 feet)	X				
3.1.5.2	Horizontal Extended Lift Capacity (15 feet)	X				
3.1.6	Safe Working Load	X				
3.1.7	Fork Carriage					X
3.1.7.1	Fork Tines (Length)					X
3.1.7.2	Fork Tines (USAF Pallet Forks)					X
3.1.7.3	Tilt					X
3.1.7.4	Oscillating					X
3.1.7.5	Offset/Shift					X
3.1.7.6	Fork Tine Spacing					X
3.1.8	Load Chart	X				
3.2.1	Steering Modes	X				
3.2.2	Terrain			X		
3.2.3	Service Brake					X
3.2.4	Parking Brake					X
3.2.5	Longitudinal Slope					X
3.2.6	Side Slope					X
3.2.7	Fording					X
3.2.8	Speed	X				
3.2.9	Turn Radius					X
3.2.10	Ground Clearance					X
3.2.11	Axle Oscillation	X				
3.3.1	Height (Cab)					X
3.3.2	Maximum Height (CH-53K Loading)					X
3.3.3	Width					X
3.3.4	Length					X
3.4.1	Internal Air Transportability					X
3.4.2	External Air Transportability					X
3.4.3	Highway Transportation					X
3.4.4	Rail Transport					X
3.4.5	Marine Transport					X
3.4.6	Roll On/Roll Off					X
3.4.7	Lifting and Tiedown Provisions					X
3.4.8	Towing Pintle	X				
3.5.1	Visibility	X				
3.5.2	Human Factors Engineering	X				

Table 4. Requirement/Verification Cross Reference Matrix - continued

Paragraph	Title	Method				
		C	A	I	D	T
3.5.3	PPE	X				
3.5.4	Cold Weather Gear	X				
3.5.5	Chemical, Biological, Radiological, Nuclear	X				
3.5.6	Air Conditioner	X				
3.5.7	Heater	X				
3.5.8	Instrumentation	X				
3.5.9	Horn	X				
3.5.10	Backup Alarm	X				
3.5.11	Ignition	X				
3.5.12	Safety	X				
3.5.13	Rifle Mount			X		
3.5.14	Noise Limits	X				
3.5.15	Hazardous Materials	X				
3.5.16	Fire Extinguisher	X				
3.5.17	Roll-Over Protective Structures	X				
3.5.18	Falling Object Protective Structures	X				
3.6.1	Tires	X				
3.6.2	Onboard Air Compressor	X				
3.6.2.1	Pneumatic Tool Storage	X				
3.6.3	Offroad and Beach Operations	X				
3.7.1	Corrosion Protection	X				
3.7.2	Paint and Treatment			X		
3.7.3	Operational Decontamination	X				
3.8.1	Low Ambient Temperature (Start-up)	X				
3.8.2	Low Ambient Temperature (Operation)	X				
3.8.3	High Ambient Temperature	X				
3.9.1	Work Lights	X				
3.9.2	Road Lights			X		
3.9.3	Blackout Lighting			X		
3.10.1	24V Electrical System			X		
3.10.2	Batteries	X				
3.10.2.1	Battery Accessibility	X				
3.10.2.2	Battery Disconnect Switch	X				
3.10.2.3	Battery Cables	X				
3.10.3	NATO Receptacle			X		
3.10.4.1	EMI					X
3.10.4.2	EMI					X
3.10.4.3	EMI					X
3.11.1	Transmission	X				

Table 4. Requirement/Verification Cross Reference Matrix - continued

Paragraph	Title	Method				
		C	A	I	D	T
3.11.1.1	Transmission Oil Sample Valve	X				
3.11.1.2	Four Wheel Drive	X				
3.11.2	Engine	X				
3.11.2.1	Fuel Capacity	X				
3.11.2.2	Engine Oil Sampling Valve	X				
3.11.2.3	Engine Oil Drain	X				
3.11.3	Exhaust	X				
3.11.4.1	Hydraulic Fluid Sample Valve	X				
3.11.5	Military Petroleum, Oil, and Lubricants (POL)	X				
3.12.1	Maintenance Infrastructure	X				
3.12.2	Preventive Maintenance	X				
3.12.3.1	Commercial Diagnostic System				X	
3.12.3.2	USMC Diagnostic Systems				X	
3.12.3.3	Diagnostic Capabilities (Read, Write, Commands)				X	
3.12.3.4	Diagnostic Capabilities (License)				X	
3.12.3.5	Diagnostic Capabilities (Documentation)				X	
3.12.4	RAM (MTBF)	X				
3.12.5	RAM (MTTR)	X				
3.12.6	Rebuild/Replace	X				
3.12.7	Common Tools	X				
3.13.1	Identification and Marking			X		
3.13.2	Instruction Plates			X		
3.13.3	Military Load Classification Identification			X		

Table 4. Requirement/Verification Cross Reference Matrix - continued

Paragraph	Title	Method				
		C	A	I	D	T
3.11.1.2	Four Wheel Drive	X				
3.11.2	Engine	X				
3.11.2.1	Fuel Capacity	X				
3.11.2.2	Engine Oil Sampling Valve	X				
3.11.2.3	Engine Oil Drain	X				
3.11.3	Exhaust	X				
3.11.4.1	Hydraulic Fluid Sample Valve	X				
3.11.5	Military Petroleum, Oil, and Lubricants (POL)	X				
3.12.1	Maintenance Infrastructure	X				
3.12.2	Preventive Maintenance	X				
3.12.3.1	Commercial Diagnostic System				X	
3.12.3.2	USMC Diagnostic Systems				X	
3.12.3.3	Diagnostic Capabilities (Read, Write, Commands)				X	
3.12.3.4	Diagnostic Capabilities (License)				X	
3.12.3.5	Diagnostic Capabilities (Documentation)				X	
3.12.4	RAM (MTBF)	X				
3.12.5	RAM (MTTR)	X				
3.12.6	Rebuild/Replace	X				
3.12.7	Common Tools	X				
3.13.1	Identification and Marking			X		
3.13.2	Instruction Plates			X		
3.13.3	Military Load Classification Identification			X		

5.0 PACKAGING

5.1. Packaging. The packaging requirements shall be specified in the contract.

6.0 NOTES

6.1. Mission Profile. The EBFL mission profile will consist of four hours of operation (1.8 hours for traveling and 2.2 hours for lifting, loading, and unloading).

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